

PROBLEM SOLVING PROTOCOL

GoCo: structure-guided conformal calibration for error-controlled Gene Ontology annotation release

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Abstract

Computational gene-function prediction generates annotations at a scale that experimental characterization cannot match, yet evaluations using subsequently acquired experimental evidence show that substantial errors remain even among leading methods. Explicit error-controlled calibration for Gene Ontology (GO) prediction remains largely undeveloped. GO prediction is both multilabel and hierarchical, and many predictors incorporate gene–gene relationships such as co-essentiality and network proximity to improve prediction. We introduce GoCo, a structure-guided framework that calibrates how scores from an existing predictor are converted into released annotations. GoCo first certifies a global score threshold for a GO-hierarchy-aware gene-level false discovery proportion (FDP), then uses that structure to construct a richer nested release path under the same asymptotic risk-control guarantee. At the primary benchmark operating point ($\alpha = 0.10$, $\delta = 0.50$), GoCo-M, developed on Wainberg co-essentiality modules and applied without retuning to three independent studies with the same shared-evidence structure, kept mean held-out gene-level FDP at or below target while releasing 6.3–17.6% more reference-supported GO terms than Multilabel. For a second predictor structure based on gene-specific neighbourhoods rather than shared modules, we developed GoCo-N, where scores are effectively untied and ordering acts on individual calls. Developed on FunMap and validated without retuning on the STRING v12.0 experiments-only network, GoCo-N released 33.8% and 36.4% more reference-supported GO terms than Multilabel on the same refined score grid in FunMap and STRING, respectively, while keeping mean held-out FDP below target. The same pattern held under the stricter $\delta = 0.10$ setting. Thus, predictor-native biological structure can be reused at calibration—not only prediction—to release more supported annotations at a chosen gene-level error target without retraining the predictor.

Key words: Gene Ontology; gene function prediction; conformal risk control; Learn-Then-Test; co-essentiality; functional networks

1. Introduction

Computational gene- and protein-function prediction is a rapidly developing area of bioinformatics. Community assessments such as CAFA have tracked hundreds of methods over more than a decade [1; 2; 3; 4], and computational annotations now outnumber experimentally supported ones by orders of magnitude [5; 6]. Most methodological development has focused on improving or benchmarking prediction scores rather than controlling error in released

annotations. Consequently, predicted annotations enter databases, enrichment analyses and mechanistic interpretation without an accompanying error statement. Benchmark-level $F_{\max}$ (the maximum F-measure across thresholds) and area under the precision–recall curve (AUPR) summarize predictive performance rather than the error of a released annotation set [7; 8; 9; 10], and annotation sets built at $F_{\max}$ -optimal thresholds can violate GO’s consistency constraints, which rule out mutually exclusive or taxonomically contradictory

functions [11]. For downstream users, the practical question is therefore not only how well functions are ranked, but how to release useful annotations while limiting unsupported ones.

Our approach is motivated by two forms of structure in GO prediction that are directly relevant to error-controlled release. The first is the GO hierarchy: GO is a directed acyclic graph (DAG) with multiple inheritance, and under the TruePath rule an annotation to a term implies annotations to its retained ancestors [5]. The second is predictor-native gene–gene structure: predictors use co-regulation, co-essentiality, physical interaction, sequence similarity and integrated networks to improve function prediction [12; 13; 14; 15; 16; 17; 18; 19]. A genome-wide score cutoff does not by itself exploit either structure: it treats the ontology as a flat label list and cannot use predictor-native evidence to retain more supported annotations at the same error target.

Several distribution-free calibration methods can add an error statement to released annotations without retraining the predictor. Split conformal prediction guarantees marginal coverage from an exchangeable calibration sample [20]. Risk-controlling prediction sets and conformal risk control extend this idea to bounded losses that are monotone in a release threshold [21; 22]. Learn-Then-Test (LTT) further allows nonmonotone candidate sequences, returning with probability at least $1 - \delta$ only candidate policies whose risk (expected loss) is at most α [23]. These approaches have been applied in molecular biology research [24; 25; 26; 27]. Hierarchical conformal methods also construct prediction sets on trees or DAGs when each unit has one true output [28; 29]. Boger et al. [24] proposed a hierarchy-aware formulation for tree-structured labels, but the multiple-parent structure of GO is not a tree and therefore requires different treatment. For GO release, the formulation must accommodate two features: a gene can have many true terms, and GO is a DAG with multiple inheritance. We define the release error as the fraction of terms released for a gene that are unsupported by the reference. Our first layer, GoDag, tests whether the average gene-level FDP for a global threshold is within a chosen limit, with a stated bound on the probability of false certification.

Existing release rules also leave two practical gaps. First, accuracy-optimized or fixed-size rules such as an $F_{\max}$ threshold or top- k terms per gene state no gene-level error rate. Second, calibrated methods restricted to a single global threshold can leave supported predictions unreleased when the score path is coarse. Conservative concentration-bound certificates can compound this loss of yield (Results). The balance between avoiding unsupported calls and retaining supported functions depends on the application: some users may place greater weight on avoiding unsupported annotations, whereas exploratory applications may place greater value on retaining potentially informative functions. Accordingly, we evaluate release rules along two complementary dimensions: gene-level FDP for unsupported releases and reference-supported yield, a complementary measure of how many supported functions are retained rather than missed, not a second error rate. The GoCo framework addresses error control and yield in sequence: GoDag supplies the gene-level error certificate, and the second layer enriches the candidate path while retaining that certification, with the appropriate refinement determined by the predictor’s evidence structure. For shared evidence, the limitation is especially clear: genes sharing module evidence can receive the same score for a term, so thousands of gene–term pairs tie exactly and no threshold can split such a block—it admits all of it, possibly above target, or none, leaving the budget unspent

(Results). But the predictor has already exposed which genes in the block share which evidence, and GoCo turns that into a nested family of release policies certified by one test.

Our contributions are fourfold. First, we define a hierarchy-aware gene-level FDP estimand and its global-threshold certificate, GoDag. Second, we construct candidate release paths for predictors that can identify the evidence behind each call; Proposition 1 and Supplementary Corollary S1 establish asymptotic finite-population fixed-sequence control for any ordering learned from the ranking fold. Third, Supplementary Lemma S6 decomposes the gain into granularity and ordering terms, clarifying how the candidate path should match the predictor’s evidence structure. Fourth, we develop two structure-specific implementations: GoCo-M for evidence shared between genes, developed on Wainberg and evaluated without retuning on Sanger Project Score, Project DRIVE and the HAP1 genetic-interaction map; and GoCo-N for per-gene neighbourhood evidence, developed on FunMap and validated without retuning on STRING. We also provide a gene-level case study of the released annotations. Code, principal analysis inputs and split-level results for the primary analyses are public, together with aggregate summaries for the secondary comparator panel.

2. Materials and methods

2.1. Step 1: define the annotation release problem

GoCo leaves the upstream predictor unchanged. It requires (i) a score s_{ig} for each gene i and GO term g and (ii) the evidence behind each candidate call, such as annotated module members or network neighbours. A call is a proposed gene–GO annotation. A release policy π specifies which calls to retain, giving each gene a set $S_i(\pi)$. A global threshold uses $S_i(\lambda) = \{g : s_{ig} \geq \lambda\}$; GoCo also considers more selective policies (Figure 1).

The Direct reference contains annotations recorded before adding their GO ancestors. TruePath closure adds the retained ancestors to give the acceptable set A_i^{TP} . The gene-level false discovery proportion is

$$L_i(S_i) = \begin{cases} 1 - \frac{|S_i \cap A_i^{\text{TP}}|}{|S_i|}, & |S_i| > 0, \\ 0, & |S_i| = 0. \end{cases} \quad (1)$$

Closure applies only to the reference; prediction scores are not propagated to ancestors. “Unsupported” means absent from this frozen reference, not biologically false. Each gene receives equal weight: one unsupported call among one released term and 20 among 20 both give $L_i = 1$; one among 20 gives $L_i = 0.05$. Releasing no terms (abstention) gives zero loss [21; 22]. We report supported yield separately because FDP does not penalize withheld functions.

2.2. Step 2: separate ranking, certification and evaluation

Each of 100 seeded random permutations divides a fixed gene panel into a ranking fold $\mathcal{T}$ (10%), a certification fold $\mathcal{C}$ (60%) and an evaluation fold $\mathcal{E}$ (30%). Ranking labels build the candidate policies; certification labels select a policy; evaluation labels assess its performance. For GoCo, the certified risk is the average FDP over the non-ranking pool $\mathcal{P} = \mathcal{C} \cup \mathcal{E}$, of size N :

$$R(\pi) = \frac{1}{N} \sum_{i \in \mathcal{P}} L_i(\pi), \quad L_i(\pi) = L_i(S_i(\pi)). \quad (2)$$

GoCo workflow

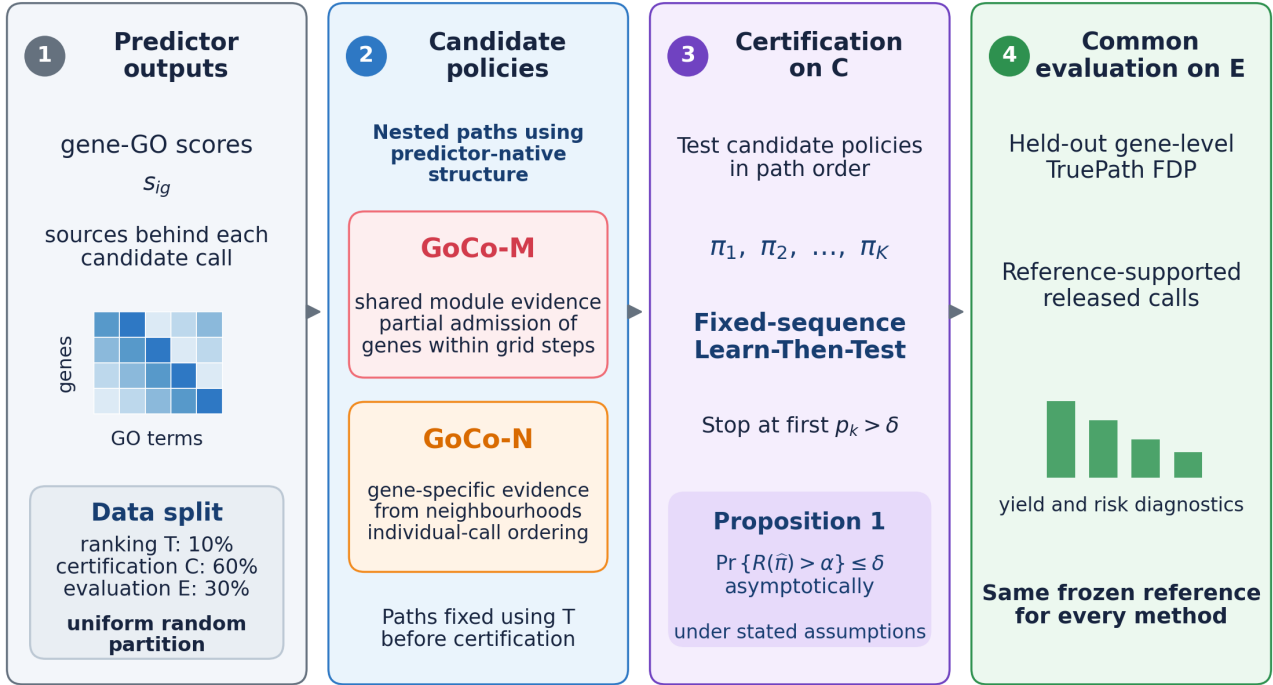


Fig. 1. Overview of GoCo and its two structural instantiations. Module analyses use 70% calibration for global methods and 60% certification for GoCo-M; neighbourhood analyses certify all methods on the same 60% fold.

Alt text: Workflow of GoCo. Predictor scores and evidence define two nested candidate-policy families: GoCo-M for shared module evidence with tied scores and GoCo-N for gene-specific neighbourhood evidence. Candidate policies are certified in fixed sequence on the certification fold, and held-out TruePath FDP and reference-supported calls are evaluated on the evaluation fold.

The target risk is $\alpha = 0.10$, with sensitivity analyses at 0.05 and 0.20. Once a candidate policy is fixed, its pool losses are constants. Conditional on the ranking fold, certification genes are sampled uniformly without replacement from that pool. This sampling design supports inference without assuming independent gene losses, an important distinction under biological dependence [30; 31]. The guarantee concerns the fixed panel and GO snapshot, not new genes outside the panel.

Module-family global methods have no ranking step and calibrate on $\mathcal{T} \cup \mathcal{C}$ (70%); their certified pool is the whole panel. All neighbourhood-family methods certify on the same 60% fold. These differences are retained when interpreting comparisons. Seeds, rounding and sample sizes are in Supplementary Note S2.1.

2.3. Step 3: construct policies from the predictor's evidence

GoDag, the first layer of GoCo, selects one global threshold using the TruePath loss. GoCo adds policies to exploit evidence already exposed by the predictor. A candidate path is a sequence of annotation sets that only grow as policies are visited. The entire path must be fixed using ranking labels and frozen inputs before certification.

Shared module evidence: GoCo-M. Many genes can receive identical scores because they share evidence. A threshold must admit or exclude an exact tie as a whole. GoCo-M keeps the predictor's threshold grid but inserts policies that admit

only some affected genes within a grid step. A logistic model trained on ranking-fold calls estimates each call's probability of being unsupported. Features describe the original score, term frequency and the reliability of its supporting sources. For each source, reliability is learned from how often the calls it supports for ranking-fold genes are reference-unsupported. Source rates are shrunk toward the ranking-fold average to stabilize estimates from small counts. Only ranking-fold labels are used as training outcomes; frozen predictor scores remain inputs.

Affected genes are ordered by predicted FDP increase per predicted supported term. The ranking fold determines successive admission cutoffs; deterministic identifier hashes resolve ties. This is a practical ordering heuristic, not an optimality claim. The complete source attribution, shrinkage formula, model, insertion criteria and cutoffs are retained in Supplementary Notes S2.3 and S4.1, with constants in Supplementary Table S11.

Gene-specific neighbourhood evidence: GoCo-N. FunMap and STRING assemble evidence separately for each gene. Their scores have few substantial ties in the operating range, so a refined global grid provides a stronger threshold baseline. GoCo-N instead orders individual calls by their predicted unsupported probability, admitting the most promising first. This creates a separate nested path, not merely GoCo-M at a finer resolution.

We chose these implementations on Wainberg and FunMap, respectively, then applied them without retuning to external studies. The choice uses the predictor’s evidence structure and score multiplicities, not evaluation outcomes. It is not universally optimal; an alternative that chooses between paths using ranking labels alone is assessed in Supplementary Note S3.6.

2.4. Step 4: certify the selected release policy

For candidate π_k , let $\hat{R}_k$ be its certification-fold mean FDP and $\hat{\sigma}_k$ its standard deviation, calculated with divisor $m = |\mathcal{C}|$. We retain the one-sided normal test used in the released Multilabel implementation [24]:

$$p_k = \Phi\left(\frac{\hat{R}_k - \alpha}{\max(\hat{\sigma}_k, 10^{-6})/\sqrt{m}}\right), \quad (3)$$

where Φ is the standard normal distribution function. Fixed-sequence Learn-Then-Test [23] tests candidates in their prespecified order, stops at the first $p_k > \delta$ and returns the preceding policy $\hat{\pi}$. If the first candidate fails, it abstains; if all pass, it returns the last. For module-family global methods, the mean, standard deviation and sample size use the 70% calibration fold instead.

Certification means statistical control of the chance of releasing a policy whose pool risk exceeds α , not just estimating its FDP. Under the stated finite-population variance and sampling conditions, that chance is asymptotically at most δ for $0 < \delta \leq 0.50$. Any ordering learned solely from the ranking fold and frozen inputs is covered. The finite-population central limit argument [32; 33], assumptions and formal statements are in Proposition 1, Supplementary Corollary S1 and Supplementary Note S1.6. Replacing the normal test with Hoeffding’s bound gives finite-sample control on the same path, usually with lower yield [34].

Proposition 1 (Conditional fixed-sequence risk control) *Let the candidate path satisfy Supplementary Definition S1, let $\delta \in (0, \frac{1}{2}]$, and let k^* be the first index with $R(\pi_{k^*}) > \alpha$ (if none exists no error is possible). Consider a sequence of such problems in which $m \rightarrow \infty$, $N - m \rightarrow \infty$ and $m/N \rightarrow f \in [0, 1)$, and suppose the pool variance $S^2(\pi_{k^*}) = (N - 1)^{-1} \sum_{i \in \mathcal{P}} \{L_i(\pi_{k^*}) - R(\pi_{k^*})\}^2$ stays bounded away from zero, and write $z_\delta = \Phi^{-1}(\delta)$. Then, conditional on $\mathcal{T}$,*

$$\limsup \Pr\{R(\hat{\pi}) > \alpha \mid \mathcal{T}\} \leq \Phi(z_\delta / \sqrt{1 - f}) \leq \delta,$$

with equality in the second inequality only when $\delta = \frac{1}{2}$ or $f = 0$. The bound holds for every $\mathcal{T}$ -measurable ordering of the affected genes within each block, and unconditionally when the variance condition holds uniformly in $\mathcal{T}$.

Proof idea. If the returned policy is invalid, the first invalid candidate on the prespecified path must have passed its test. False certification is therefore bounded by that single test, without assuming independence among candidate tests or monotonicity of FDP along the path. See Supplementary Note S1.6 for the full proof.

We report $\delta = 0.50$ to match the released Multilabel default and $\delta = 0.10$ as a stricter analysis. At 0.50, Equation (3) reduces to $\hat{R}_k \leq \alpha$ and its formal false-certification bound is permissive (0.50). Held-out FDP can exceed α even when pool risk does not; these are different events. The exact relationship and stronger asymptotic bound under sampling without replacement are in Supplementary Remark S2.

2.5. Data, comparators and reporting

Wainberg’s co-essentiality predictor uses leave-one-gene-out GO enrichment of overlapping modules [19]. Its reconstructed scores match the archived file to numerical precision. Sanger Project Score [35] and Project DRIVE [36] supply CRISPR and RNAi evidence to select among these modules without GO-based selection; HAP1 uses published genetic-interaction clusters [37]. Panels contain 15,794, 14,885, 6,283 and 15,020 genes, respectively. All four use the GO release of 1 November 2019. For these module datasets, the reference includes all evidence codes, removes NOT-qualified annotations and root terms, and uses *is.a* and *part.of* for TruePath closure. FunMap [38] and STRING v12.0 experiments-only [39] use target-excluded top-50 random-walk neighbourhoods and GO release 2024-09-08. Their full construction and leakage checks are in Supplementary Note S3; module reconstruction and reference details are in Supplementary Note S4.3.

All methods use the same frozen scores, split streams and held-out TruePath reference within each dataset. Multilabel uses the Direct loss, whereas GoDag uses TruePath and GoCo also uses the predictor’s evidence. Generic Direct-loss comparators use Hoeffding [34], Hoeffding–Bentkus [40], an independent-observation normal approximation, empirical Bernstein [41], McDiarmid [42], Janson [43], network heteroskedasticity and autocorrelation consistent (HAC) variance [44], or Platt [45] and isotonic [46] calibration. Graph/HAC assumptions are not established by our sampling design, so those are working-model comparisons. Platt and isotonic calibration provide no guarantee for the target risk. The method inventory is retained in Supplementary Table S12.

We report mean held-out gene-level FDP, supported calls and genes, total calls, and pool-level failure and held-out exceedance frequencies over 100 splits. A supported call is a released gene–GO pair confirmed by the frozen reference. Split variation reflects random partitioning of this panel, not sampling new biological populations; frequency estimates have Monte Carlo standard errors up to 0.05. Supplementary Note S2.7 defines all quantities and the handling of abstention.

3. Results

3.1. Tied scores leave supported annotations unreleased

Wainberg contains 2,470 gene–GO calls with the identical score 789.650. Its threshold step from 800 to 775 raises full-panel TruePath risk from 0.0924 to 0.1398 (Figure 2). No threshold can divide the dominant tie, so GoDag stops at 800 in all 100 splits despite a target of 0.10. Partial admission provides intermediate policies. Random admission reaches the target at about 16% in a descriptive full-panel interpolation; GoCo’s informed ordering allowed certification at admission settings of 35–40% in 89 splits. The interpolation is not itself a certificate. Supplementary Lemma S6 separates the contributions of finer admission steps and informative ordering without claiming guaranteed yield improvement. Across the certified module-family blocks, the ordering statistic’s Spearman correlation with realised loss increments was 0.22–0.73 on non-ranking genes (Supplementary Table S4).

3.2. Shared module evidence: GoCo-M

At $\alpha = 0.10$, $\delta = 0.50$, GoCo-M’s mean held-out FDP was 0.0964, 0.0980, 0.0971 and 0.0943 in Wainberg, Sanger, DRIVE and HAP1 (Table 1; Figure 3). Supported yield increased by

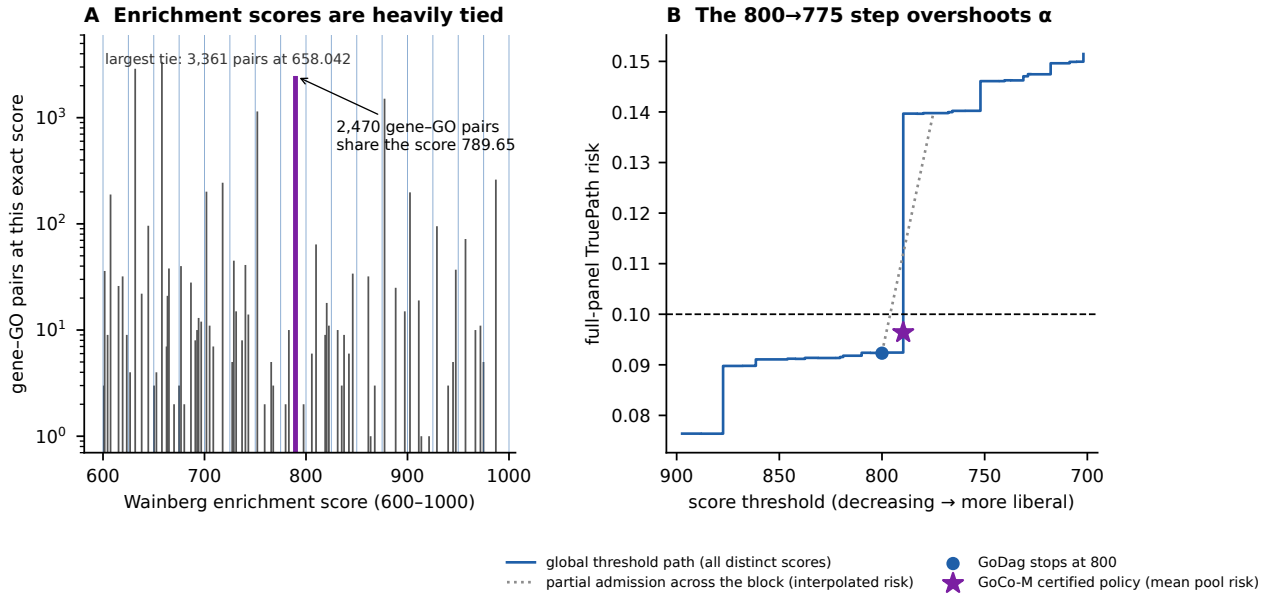


Fig. 2. Score ties and the global-threshold ceiling in Wainberg. (A) Exact-score multiplicities between 600 and 1000. (B) Full-panel TruePath risk across distinct thresholds, highlighting the 800 $\rightarrow$ 775 step that crosses the $\alpha = 0.10$ target and the partial-admission path within that step.

Alt text: Two-panel illustration of score ties in Wainberg. Panel A shows large multiplicities of identical enrichment scores, including 2,470 gene-GO pairs at score 789.65. Panel B shows the full-panel TruePath risk jumping across the 0.10 target between thresholds 800 and 775, whereas partial admission interpolates within the tied block.

8.7%, 14.3%, 17.6% and 6.3% relative to Multilabel. GoDag improved on Multilabel when TruePath changed the selected threshold; GoCo-M added supported calls in all four datasets. Comparisons with GoDag assess the full procedure, not the path alone, because their certification folds differ (60% versus 70%). Gains persisted at $\delta = 0.10$: 7.8%, 7.7%, 15.6% and 6.3%, respectively (Supplementary Table S1; Supplementary Figure S1).

No generic Direct-loss comparator exceeded GoDag in supported yield: the concentration-bound members annotated fewer genes at a realised FDP below target, and at $\delta = 0.50$ the normal-type rules coincide with Multilabel exactly (Supplementary Remark S2). GoCo-M's distinguishing feature is a richer set of candidate policies, not a sharper bound on the same threshold path.

3.3. Confidence and error-target sensitivity

GoCo-M exceeded the pool-risk target in 5–26% of splits at $\delta = 0.50$, below that setting's permissive formal bound. At $\delta = 0.10$, pool failure occurred in zero or one split per module dataset. These frequencies are consistent with, but too coarse to assess precisely, asymptotic upper bounds near 0.013 for GoCo and 0.010 for global methods. Held-out exceedance remained more frequent (Supplementary Tables S1 and S3).

At $\alpha = 0.05$ and 0.20, mean held-out FDP remained below target for the principal methods, and GoCo-M matched or exceeded Multilabel's supported yield (Figure 4; Supplementary Tables S13 and S2). Gains were largest where a grid step crossed the target sharply. The zero gain in DRIVE at $\alpha = 0.20$, $\delta = 0.50$ arose because both methods reached the grid's reporting floor.

3.4. Yield and biological interpretation

Only about 8–12% of held-out genes received calls in the module analyses. The gene-averaged target includes abstentions and controls neither the FDP among annotated genes alone nor the pooled fraction of unsupported calls. Additional yield therefore comes with additional unsupported releases. At $\delta = 0.10$, GoCo-M released 1,896 versus GoDag's 1,614 calls in Wainberg and 1,252 versus 847 in HAP1; the supplement reports their supported fractions (Supplementary Tables S3 and S6).

In Wainberg split 0 at $\delta = 0.50$, GoCo-M released 351 held-out calls that GoDag did not, over 158 genes; 37 (10.5%) were reference-supported. All four interleukin-27/35 signalling calls added for IL6ST and JAK1 in an interleukin-6 signal-transduction module were supported. CDKN1A belonged to a module mixing a regulator with its targets; none of the seven transferred TP53 terms was supported. These examples illustrate both useful evidence sharing and its limits. N4BP1's added Toll-like receptor signalling annotation was absent from the frozen reference despite published biological support [47]. The full case study is retained in Supplementary Note S2.9 and Table S5.

3.5. Gene-specific neighbourhood evidence: GoCo-N

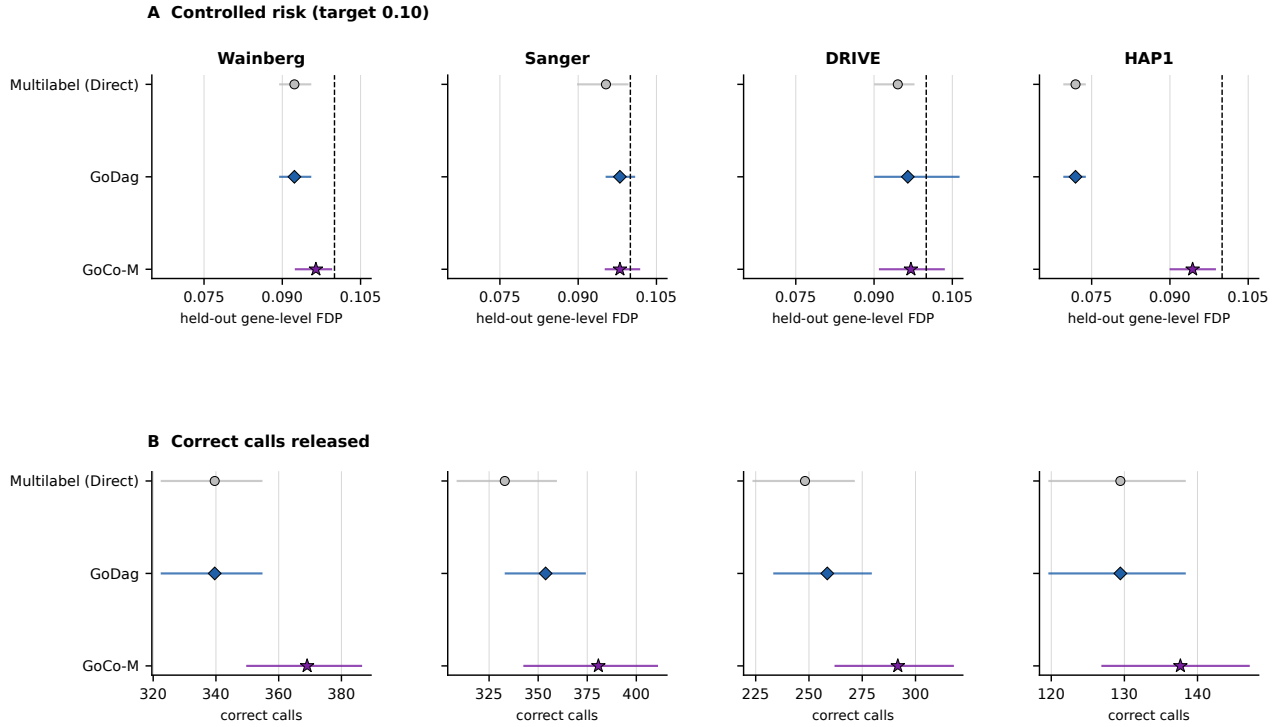
Developed on FunMap and validated without retuning on STRING's experiments-only network, GoCo-N increased supported calls by 33.8% and 36.4% relative to Multilabel on the same refined score grid. Mean held-out FDP was 0.0921 and 0.0944 (Table 2). Compared with refined-grid GoDag, it added 342 and 246 supported calls. The stricter $\delta = 0.10$ gains over Multilabel were 34.0% and 35.4%. Supplementary Table S8 compares yield with the best global threshold at matched realised risk.

Table 1. Module-structure comparison of mean held-out FDP and supported yield at $\alpha = 0.10$, $\delta = 0.50$, over 100 splits. Wainberg is the development data set; Sanger, DRIVE and HAP1 are external studies with the same shared-evidence structure.

Method	Wainberg ($n_{\mathcal{E}}=4,738$)			Sanger ($n_{\mathcal{E}}=4,465$)			DRIVE ($n_{\mathcal{E}}=1,885$)			HAP1 ($n_{\mathcal{E}}=4,506$)		
	Gene FDP	Genes with a correct call	Correct calls	Gene FDP	Genes with a correct call	Correct calls	Gene FDP	Genes with a correct call	Correct calls	Gene FDP	Genes with a correct call	Correct calls
<i>Direct loss, one global threshold (ontology-blind)</i>												
Multilabel	0.0923	185.7	339.6	0.0953	155.4	333.0	0.0945	96.1	248.1	0.0719	77.0	129.4
Hoeffding/LTT	0.0914	182.6	333.3	0.0876	148.1	308.0	0.0864	84.2	206.1	0.0719	77.0	129.4
Hoeffding-Bentkus	0.0923	185.7	339.6	0.0916	150.9	316.4	0.0923	91.8	233.9	0.0719	77.0	129.4
IID-normal	0.0923	185.7	339.6	0.0953	155.4	333.0	0.0945	96.1	248.1	0.0719	77.0	129.4
Empirical Bernstein	0.0919	183.6	335.7	0.0894	149.3	311.5	0.0882	86.0	213.1	0.0719	77.0	129.4
McDiarmid (binary incidence)	0.0804	166.1	291.8	0.0777	142.2	288.6	0.0782	78.4	183.3	0.0719	77.0	129.4
Janson dependency graph	0.0775	162.1	281.0	0.0769	141.7	287.0	0.0746	76.0	175.7	0.0719	77.0	129.4
Network-HAC ($b = 1$)	0.0923	185.7	339.6	0.0953	155.4	333.0	0.0945	96.1	248.1	0.0719	77.0	129.4
Platt calibration	0.0908	182.3	332.1	0.0949	155.6	334.4	0.0914	91.7	233.6	0.0719	77.0	129.4
Isotonic calibration	0.0919	184.8	337.6	0.0956	156.5	337.8	0.0917	92.0	235.3	0.0719	77.0	129.4
<i>TruePath loss, one global threshold (GO hierarchy)</i>												
GoDag	0.0923	185.7	339.6	0.0980	160.9	353.8	0.0965	98.7	258.6	0.0719	77.0	129.4
<i>TruePath loss, structure-guided partial admission (GO hierarchy and module-shared evidence)</i>												
GoCo-M	0.0964	197.1	369.1	0.0980	166.9	380.7	0.0971	105.7	291.7	0.0943	80.8	137.7

Gene FDP: mean held-out TruePath false discovery proportion. Correct calls: reference-supported gene–GO pairs. Genes with a correct call: genes with at least one supported release. All global methods use the same 25-unit grid and 70% calibration; GoCo-M uses 10% ranking and 60% certification. Platt/isotonic rules are δ -free and have no risk guarantee. Normal-type rules coincide at $\delta = 0.50$ on the same path, loss and calibration sample (Supplementary Remark S2).

$\alpha = 0.10$, $\delta = 0.50$: mean over 100 splits (marker) and interquartile range (bar)

**Fig. 3.** Primary comparison of Multilabel, GoDag and GoCo-M at $\alpha = 0.10$, $\delta = 0.50$: held-out gene-level TruePath FDP and reference-supported calls over 100 splits.

Alt text: Primary comparison across Wainberg, Sanger, DRIVE and HAP1. The upper row shows mean held-out gene-level TruePath FDP and interquartile ranges for Multilabel, GoDag and GoCo-M relative to the 0.10 target. The lower row shows reference-supported released gene–GO calls, with GoCo-M releasing the most in each data set.

The structural choice is useful but not uniformly best. At $\alpha = 0.10$, $\delta = 0.10$, GoCo-N lost 142 supported terms relative to GoCo-M on Wainberg, while it performed better on Sanger and HAP1. A ranking-fold-only selection rule agreed with the held-out winner in 69–100% of splits across predictors (Supplementary Note S3.6). This remains a refinement rather than the primary prespecified analysis.

3.6. Comparison with accuracy-based release rules

A validation $F_{\max}$ threshold and one top-scoring term per gene produced gene-level FDPs of 0.18–0.99 and 0.17–0.98, respectively. Both exceeded 0.10 in every split of every predictor (Supplementary Table S10). They can recover more supported functions by releasing more calls, but provide no way to certify the accompanying gene-level error rate.

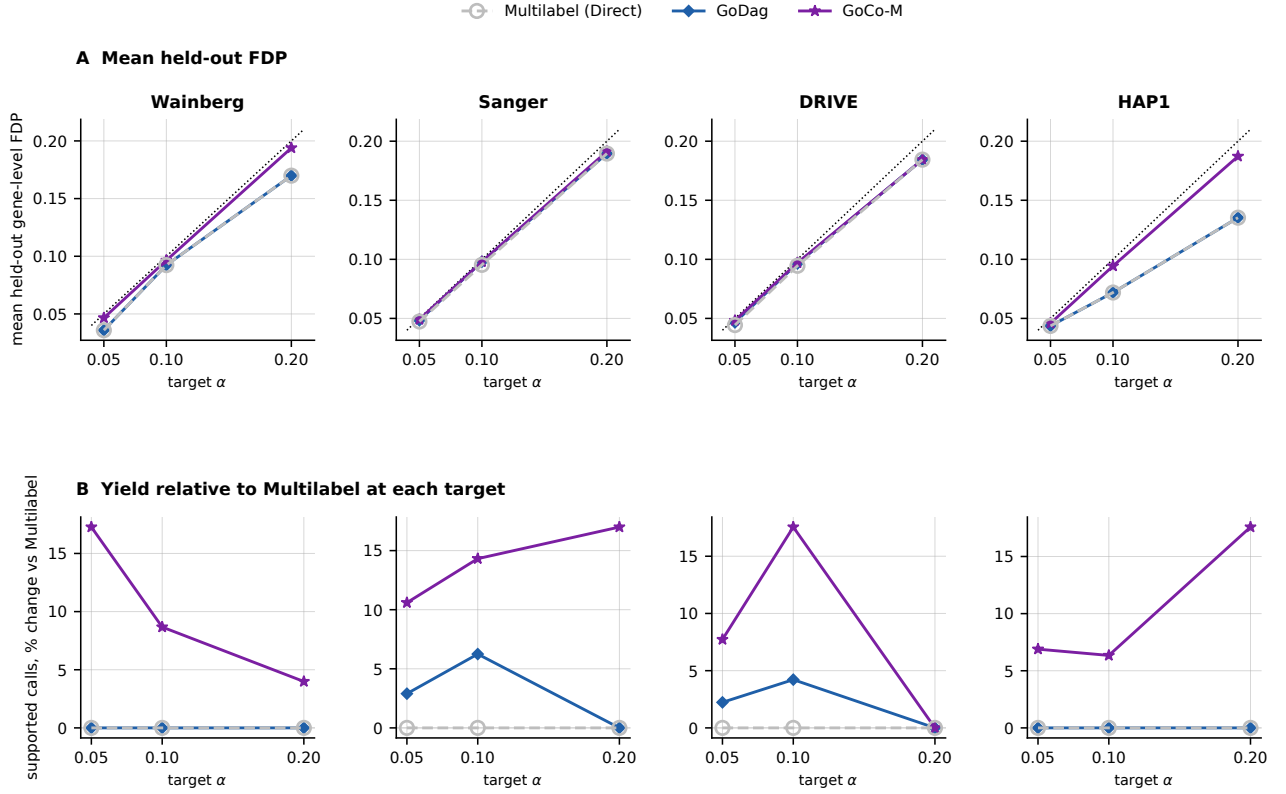


Fig. 4. Error-target sensitivity at $\delta = 0.50$. Top: mean held-out gene-level FDP; the dotted line is the target. Bottom: supported-call gain relative to Multilabel. Method colours match Figures 2 and 3 and Supplementary Figure S1. Open grey circles keep Multilabel visible where it overlaps GoDag. Full values are in Supplementary Table S13.

Table 2. Neighbourhood-structure comparison at $\alpha = 0.10$, $\delta = 0.50$, over 100 splits. FunMap is the development data set; the STRING v12.0 experiments-only network provides the external validation.

Method	FunMap ($n_{\mathcal{E}}=3,095$)			STRING ($n_{\mathcal{E}}=2,417$)		
	Gene FDP	Correct calls	Genes with a call	Gene FDP	Correct calls	Genes with a call
<i>Direct loss, one global threshold on the refined grid (ontology-blind)</i>						
Multilabel	0.0888	1245	734	0.0899	885	432
Hoeffding/LTT	0.0820	1182	699	0.0820	826	405
Hoeffding–Bentkus	0.0855	1213	717	0.0855	857	419
IID-normal	0.0888	1245	734	0.0899	885	432
Empirical Bernstein	0.0835	1193	706	0.0831	837	410
Platt calibration	0.0647	1061	615	0.1003	989	487
Isotonic calibration	0.0824	1184	701	0.0874	869	424
<i>TruePath loss, one global threshold (GO hierarchy)</i>						
GoDag	0.0901	1261	742	0.0828	828	406
GoDag, refined grid	0.0986	1324	783	0.0978	962	472
<i>TruePath loss, structure-guided ordering of calls (GO hierarchy and neighbourhood evidence)</i>						
GoCo-N	0.0921	1666	1110	0.0944	1208	837

Gene FDP: mean held-out TruePath false discovery proportion. Correct calls: reference-supported gene–GO pairs. Genes with a call includes unsupported releases. Direct-loss rules use the refined grid; GoDag is shown on both coarse and refined grids. All methods certify on the same 60% fold. Platt/isotonic rules have no risk guarantee. McDiarmid, Janson and network-HAC variants were not evaluated here because their graph/variance specification was not prespecified.

4. Discussion

GoCo addresses a practical problem: deciding which annotations to release from an existing predictor at a stated error target. GoDag supplies a hierarchy-aware global-threshold baseline; GoCo uses evidence behind calls to improve supported

yield. The protocol separates ordering from certification, allowing users to compare release policies without retraining their predictor.

The target must match the user’s purpose. GoCo’s gene-averaged FDP follows risk-control formulations used in conformal calibration and enzyme annotation [22; 25]. It is

not pooled call-level FDR [48; 49] or a guarantee for each gene or subgroup [50; 51]. The pooled-call comparator in Supplementary Table S6 releases almost nothing at the same numerical target, but addresses a different estimand; pair-level exchangeability is not established by our gene-level split. Using side information to improve yield also connects GoCo to covariate-adaptive testing [52; 53] and smoothed conformal prediction [54], with different targets and constructions.

GO is incomplete and open-world: missing annotations are not evidence of absent function [55; 56; 57]. Functional relevance also varies by tissue, cell state, development, species and experimental condition. Supported yield and reference-based FDP should therefore be interpreted together, not as complete measures of biological correctness. Evidence sources must exclude the target gene's own annotations to avoid circular support. Supplementary Note S2.7 records the remaining label information already present in frozen enrichment scores.

Limitations and use beyond these predictors

The guarantee is asymptotic, requires nonvanishing variance at the first invalid candidate and applies to this fixed panel and reference. The primary $\delta = 0.50$ setting is a benchmark convention, not a strong deployment confidence level. Sanger, DRIVE and HAP1 contribute independent studies but share Wainberg's gene universe and reference; they are not independent predictor families. The better path cannot yet be predicted reliably from structure alone.

GoDag needs only scores. GoCo additionally needs interpretable evidence behind individual calls. Protein complexes and co-regulation clusters are possible shared sources; homology hits [9; 12], embedding neighbours [58; 24; 25] and integrated networks [17] are possible gene-specific sources. These extensions require empirical validation, not simply adoption of the structural default.

5. Conclusions

GoCo combines an unchanged predictor, GO-aware error assessment and structure-guided release policies. Across six applications, the structure-specific GoCo implementations increased supported yield while keeping mean held-out gene-level FDP at or below target; the four external applications were evaluated without retuning. The practical contribution is a reproducible workflow for comparing annotation release rules, interpreting their error statements and retaining more reference-supported functions.

Key Points

- Accuracy-optimized and fixed-size release rules provide no gene-level error certificate; FDP ranged from 0.17 to 0.99 across these applications.
- GoCo combines GO-aware calibration with evidence already exposed by an unchanged predictor.
- An asymptotic certificate controls pool-average gene-level FDP for paths built without certification or evaluation labels, not pooled-call FDR or each gene's error.
- GoCo-M increased supported yield by 6.3–17.6% over Multilabel across four module datasets at the primary operating point.

- GoCo-N increased supported yield by 33.8% and 36.4% in FunMap and STRING against Multilabel on the same refined score grid.

Supplementary data

Supplementary material accompanies this article online.

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Conflict of interest

None declared.

Data availability

No new experimental data were generated in this study. The reference implementation of both families (`goco_rerun.py` and `goco_learned.py` for the module family; `goco_second_family.py` and `run_goco_second_family.py` for the neighbourhood family), the measurability and verification tests, the table and figure scripts, the explicit fold assignments of all 100 splits, the tie-multiplicity counts, the principal split-level analysis outputs and the aggregate summaries used for the secondary comparator panel are openly available in the GitHub repository <https://github.com/ChongWangStat/ConformalGoCo>. The released software and accompanying reproducibility materials are archived on Zenodo under the stable concept DOI <https://doi.org/10.5281/zenodo.22677009> [59].

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